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Studies on floristic diversity of Sacred Grove of Potka Block, Purbi Singhbhum, Jharkhand, India

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Abstract- Sacred Grove is a traditional method of community-driven conservation, acknowledged as regions of cultural and religious significance to local populations. Sacred groves play a crucial ecological role by preserving biodiversity, safeguarding water sources, and preventing soil erosion. Groves serve as vital biodiversity hotspots, containing a high richness of flora and fauna. They play a key role in preserving rare and endangered species. Jharkhand is known as the land of forest with its vibrant tribal communities. The sacred groves of Jharkhand known as Jaherthan plays a crucial role in preserving floristic diversity, maintaining tribal culture, and promoting conservation efforts. The floristic diversity including distribution, abundance and plant species richness of two sacred groves Harina Jaherthan and Edeldih Jaherthan of Potka block, Purbi Singhbhum district of Jharkhand, India, was studied along with its traditional significance. The study documents taxonomical identification of 20 major plant species, including trees, shrubs, climbers, and herbs, each fulfilling different ethnobotanical roles present inside the grove. It also examines how external influences such as urban growth impact traditional knowledge systems, leading to the erosion of long-standing ecological practices. The findings underscore the need for continued conservation efforts to preserve both biodiversity and tribal culture through sacred grove protection.

Keywords: Sacred Grove, Traditional, Floristic Diversity, Biodiversity Conservation, Jharkhand

INTRODUCTION

Globally, Sacred Groves are seen as a traditional method of conservation led by communities, valued for their cultural and religious significance to the local populations. Sacred groves are areas of forest or natural vegetation that are protected and conserved by local communities, often tribal or indigenous groups due to their spiritual, cultural or religious significance. Sacred groves in India are important repositories of biodiversity and play a crucial role in preserving indigenous knowledge and traditions. Sacred Groves-1) it is the place where the deity resides, 2) Temple Grove is created around a temple and conserved, 3) Grove around the burial or cremation grounds.[1] Sacred groves are considered to be a sanctuary for rare, endangered, threatened, and endemic species, which remain untouched and safeguarded by local communities due to their reverence for the forest deities. These groves are safeguarded by different ways like open sacred groves, concrete boundary, bamboo boundary and stone boundary.

Sacred Grove of Jharkhand:

Jharkhand is recognized as one of India's biodiversity-rich states due to its unique origin and varied geographical and climatic conditions. It is renowned for its tribal

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Figure 1: Sacred Grove

communities, abundant mineral deposits, and extensive forest resources. Jharkhand, an eastern state of India, was carved out of the southern part of Bihar State on 15 November 2000. Jharkhand shares its border with the states of Bihar to the north, and Chhattisgarh to the west, Orissa to the south, and West Bengal to the east. It has an area of 79,714 km^2 or 30,778 sq. mi (79,710 km^2). The name "Jharkhand" means "The Land of Forests". Jharkhand state is well known for its variegated tribal population. It holds the 6th rank in terms of scheduled tribes' population in India with total of 32 scheduled tribes. The Munda, Santhal, Oraons, Ho, Bhumij are the major tribes in the region.[2] The tribals of Jharkhand worship their sacred groves known as sarnas or jaherthan. A total of 29 sacred groves has been documented so far. A sarna is a cluster of trees where the tribals would worship in various occasions. Sacred groves were analysed across different regions of Jharkhand and highlighted the unique species composition, endemic and rare plants and the ecological service provided by these ecosystems.[3]

LITERATURE REVIEW

Sacred groves have been studied in many African countries, including Kenya, Tanzania and South Africa and most West African countries: Benin, Ghana, Guinea Bissau, Nigeria, Sierra Leone, Senegal and Togo.[4,5] 29 plants were identified, four of which are listed as threatened, and 23 wild animals with one species also threatened, in the sacred groves. The sacred groves are owned by the communities, but held in trust by the chiefs. Management of the groves is by the tindanas in collaboration with the chief and elders. Traditional management systems in use are taboos, restrictions and bye-laws.[6]

It has more documented sacred groves than any other country in the world. Out of the country's 28 states, 19 are reported to contain sacred groves.[7] It has been estimated that between 100,000 and 150,000 sacred groves exist in India[8], although some of these groves are too small to contribute significantly to biodiversity conservation. The sacred groves of Garhwal Himalaya, Uttarakhand was studied and the plant species that is considered sacred in the Himalaya are *Ficus benghalensis*, *Ficus religiosa, Ocimum sanctum*, *Cynodon dactylon*, *Mangifera indica, Astromonium* spp, *Azadirachta indica*, etc. were recorded.[9] The cultivated and wild plant species assemblages of 69 sacred sites in the megacity of Bengaluru, India, was analysed in relation to biological and cultural features, and parameters related to the urban matrix and type of sacred sites (temple vs. katte). They found a dominance of native species in the cultivated and spontaneous species pools (121 species in total), with *Ficus religiosa* and *Azadirachta indica* as most frequently planted species. The occurrence of different groups of cultivated and spontaneous (i.e., non-cultivated) plants in urban sacred sites, with a focus on culturally important plant species that may functions as indicators of cultural ecosystem services.[10]

Sacred groves in Jharkhand, India, are traditionally preserved forest patches that hold significant ecological and cultural value. Understanding the floristic diversity of these sacred groves is crucial for biodiversity conservation and for maintaining the ecological balance in the region. Various grove temples of the state of Jharkhand were studied that is created or worshipped by different communities and tribes. The paper also discussed the priest or shamans who is non Sanskritic and non-Brahmanical conduct the rituals for votary in these grove temples.[11] The ecological study of the vegetation inside the sacred groves of Santhal of Dumka block. The study is based on analytical characters.[12] The documentation of floristic diversity in Jharkhand's sacred groves is an ongoing process, with studies focusing on identifying and documenting the plant species within these groves. To date, only 29 sacred groves have been recorded in Jharkhand, yet there remains a significant number of these groves that are still unexplored and undocumented.

MATERIAL & METHODS

Study Area:

A survey was carried out in the Potka block of East Singhbhum district of Jharkhand to study the floristic

diversity of different sacred groves present. The present paper documents the floristic diversity of Harina Jaherthan (Magad Buru Disom Jaherthan) and Edeldih Jaherthan (Disom Jaher Edeldih), also highlighting their traditional role in biodiversity conservation.



Figure 2: Map of Jharkhand



Figure 3: Google earth map of studied sacred groves in Purbi Singhbhum (East Singhbhum), Jharkhand.

The sacred grove was selected based on the plant diversity. A survey was conducted using a semi-structured questionnaire with the village head (Pahaans) and other knowledgeable members of the village about the grove. The geographical information was noted using the google earth software tool. The plant species present inside the grove was identified and recorded. The ethnobotanical role and importance of each plant species was also recorded. Then the external influences such as destruction and urban growth were observed and recorded.

RESULTS

The sacred groves of Potka block of Purbi Singhbhum district of Jharkhand showed a high floristic diversity. Harina sacred grove known as Magad Buru Disom Jaherthan is owned by the Bhumij community and Edeldih Jaherthan also named as Disom Jaher Edeldih is owned by the Santhal community. The taxonomical analysis showed about 40 plant species were found including trees, shrubs, herbs and climbers among which 20 major plant species were documented having cultural as well as ethnobotanical significance. Most dominating plant species was of tree species including *Shorea robusta, Madhuca longifera* and *Albizia lebbeck.* Other plant species were also documented including *Butea monosperma, Syzgium cumini, Semecarpus anacardium, Croton caudatus, Holarrhena pubescens, Dioscorea wallichii. The* analysis also includes a list of plant species with its botanical name, local name, family, habitat, purpose of use and its status according to the International Union for Conservation of Nature (IUCN). It documented 2 endangered species, 2 that are near threatened, 1 species classified as critically endangered, 7 species of least concern, and other species that have not been assessed. The unevaluated species might be vulnerable.



Figure 4: Worshiping spirit inside the grove of Harina Jaherthan.



Figure 5: Worshiping spirit of Edeldih Jaherthan

Table 1: List of major plant species documented from the Sacred Groves (Jaherthan) of Potka block, Purbi Singbhum district of Jharkhand.

Sl. No.	Botanical name	Local name	Family	Habitat	Purpose of Use	IUCN Status
1	*Abrus precatorius* L.	Gunchi	Fabaceae	Climber	Medicinal use	NE
2	*Azadirachta indica* A.Juss.	Neem	Meliaceae	Tree	Religious, Medicinal	LC
3	*Buchanania cochinchinesis* (Lour.) M.R. Almedia	Chironji	Anacardiaceae	Tree	Medicinal	CR
4	*Butea monosperma* (Lam.) Taub.	Palash	Fabaceae	Tree	Religious, Medicinal	LC
5	*Hellenia speciosa* (J.Koenig) S.R. Dutta	Keukand	Zingiberaceae	Herb	Medicinal	NT
6	*Cissus quadrangular* L.	Harjoda	Vitaceae	Climber	Medicinal	NE
7	*Cleome viscosa* L.	Hurhura	Cleomaceae	Herb	Medicine, Food	NE
8	*Diospyros melanoxylon* Roxb	Kendu	Ebenaceae	Tree	Religious, Medicinal	NE
9	*Euphorbia neriifolia* L.	Seeth	Euphorbiaceae	Shrub	Religious	LC
10	*Ficus benghalensis* L.	Bargad	Moraceae	Tree	Religious	NE
11	*Hemidesmus indicus* (L.) R. Br.	Magrabu	Asclepiadaceae	Shrub	Medicinal	EN
12	*Madhuca longifolia* (L.) J.F.Macbr.	Mahul	Sapotaceae	Tree	Religious, Medicinal	NE
13	*Plumeria rubra* L.	Gulaich	Apocynaceae	Shrub	Religious	LC
14	*Semecarpus anacardium* L.f.	Bhelwa	Anacardiaceae	Tree	Medicinal	LC
15	*Shorea robusta* C.F. Gaertn.	Sorjum	Dipterocarpaceae	Tree	Religious	NT
16	*Strychnos potatorum* L.f.	Katak	Loganiaceae	Tree	Medicinal	NE
17	*Terminalia elliptica* Willd.	Asan	Combretaceae	Tree	Religious, Medicinal	NE
18	*Tetctona grandis* L.f.	Shagwan	Lamiaceae	Tree	Religious, Medicinal	EN
19	*Waltheria indica* L.	Khar Dudhi	Malvaceae	Shrub	Medicinal	LC
20	*Zizphus oenopolia* L.Mill.	Bir Jamun	Rhamnaceae	Shrub	Medicinal	LC

*LC- Least Concern, NE- Not Evaluated, NT-Near Threatened, EN- Endangered, CR- Critically Endangered

DISCUSSION

The study indicates that the two sacred groves in the Potka block of the Purbi Singhbhum district in Jharkhand exhibit a high level of floristic diversity. The sacred grove of Harina village named as Magad Buru Disom Jaherthan (Harina Jaherthan) is owned by the Bhumij tribal community. The bhumij tribals visit their sacred grove during Haadi (Sarhul) festival to worship their spirit god and perform many cultural activities. The rituals are performed under the *Shorea robustra* tree, *Haldina cordifolia, Ficus benghalensis. The* grove has few plants which is used as traditional medicinal practices. The sacred grove of Edeldih village named as Disom Jaher Edeldih (Edeldih Jaherthan) is owned by the Santhal tribal community. They usually visit the grove during Baha festival. The rituals are performed under the hut which is surrounded by the Sal tree. The sacred grove is about 2 acre which comes under the forest area. The most dominanting plant species is *Shorea robusta* followed by *Diospyros melanoxylum* and *Termialia elliptica*. Many plant species with medicinal properties were commonly present within both the sacred groves, these include *Cheilocostus speciosus*, *Semecarpus anacardium, Bacopa monnieri, Strychnos potatorum.*

Due to external activities the sacred grove has introduced many invasive species which as degrading the native plant species of the area. The most common invasive species obsereved were *Lantana camara*, *Mikania micrantha* and *Parthenium hysterophorus* (Congress weed). The plant species within the sacred grove are never removed or used for any purpose, as each native species holds spiritual significance that has been passed down through generations by the tribal community. This traditional practice leads to the biodiversity conservation.

Threats factors for sacred grove:

Many factors that are threat to the sacred groves was observed. The study showed the invasive plant species like *Mikania micrantha*, *Lantana camara*, and *Chromolaena odorata* often overshadow the young native plants, playing a crucial role in environmental degradation. Cattle grazing was also one of the threats which degraded the area. Rapid economic development, urbanization, and industrialization lead to the clearing of groves for the construction of roads, parking lots, projects, and other infrastructure. The process of modernization is contributing to the decline of traditional knowledge concerning sacred groves and the significant ethno medicinal properties of the plant species found within them.

CONCLUSION

Sacred groves is a crucial practice for maintaining the biodiversity of our ecosystem, as they play a key role in the in-situ conservation of various endangered species. Sacred groves in Jharkhand offer unique ecosystems that preserve a rich diversity of plant species. These groves often contain remnants of original vegetation that have been protected due to religious and cultural beliefs. Documenting the floristic diversity helps to preserve the cultural heritage and the traditional knowledge associated with Sacred Grove and will be passed down to younger generations, maintaining cultural continuity. Many sacred groves contain plants with medicinal properties their documentation can help preserve knowledge about traditional remedies and potentially contribute to modern medical research The study on the floristic diversity within Jharkhand's sacred groves reveals that these regions are biodiversity hotspots, particularly abundant in trees, herbs, and medicinal plants. These ecosystems are maintained through traditional practices but are under significant threat from industrialization and deforestation. The findings highlight the need to incorporate these sacred groves into conservation management strategies to safeguard their ecological and cultural importance for sustainable future preservation.

To enhance the role of sacred groves in biodiversity conservation we have to educate local communities and visitors about the ecological and cultural importance of sacred groves. Record and preserve the traditional ecological knowledge associated with sacred groves to ensure its continuity. Encourage scientific studies on the biodiversity and ecosystem services provided by sacred groves to inform conservation strategies. By recognizing and supporting the role of sacred groves in biodiversity conservation, we can harness the power of cultural traditions to protect valuable ecosystems and promote sustainable development.

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